

WATERBURY-OXFORD, CT, USA

WMO: 725029

Lat: 41.483N

Lon: 73.133W

Elev: 726

StdP: 14.31

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 64707

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.1	8.5	-11.1	3.1	7.6	-7.0	3.9	11.1	24.8	26.3	20.9	24.8	7.3	330	0.490

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.3	87.7	73.5	84.0	71.6	81.6	69.9	75.9	83.6	74.2	80.8	72.6	78.5	6.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
73.2	126.8	79.3	72.3	122.8	78.0	71.0	117.2	76.4	39.9	83.2	38.1	80.6	36.7	78.2	81.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
19.6	17.1	15.1	-3.2	91.4	4.5	2.8	-6.5	93.5	-9.1	95.1	-11.7	96.7	-15.0	98.7	
			DB												
			WB	-2.1	78.2	5.2	1.8	-5.8	79.5	-8.9	80.5	-11.8	81.5	-15.6	82.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.0	26.0	29.5	36.0	46.8	56.3	66.0	70.8	69.3	62.4	51.5	40.9	31.5
	DBStd	17.40	11.22	9.69	8.74	8.11	6.95	6.22	5.06	5.51	6.60	7.94	8.59	8.96
	HDD50	2879	745	576	442	153	16	0	0	0	1	77	293	576
	HDD65	6341	1209	994	898	548	284	64	10	21	127	425	722	1039
	CDD50	2526	2	2	9	57	212	479	646	599	372	123	22	3
	CDD65	511	0	0	1	2	15	93	191	155	48	6	0	0
	CDH74	3595	0	0	6	36	203	658	1406	1014	244	28	0	0
	CDH80	880	0	0	1	10	48	169	365	241	43	3	0	0
Wind	WSAvg	6.4	7.1	7.2	7.8	7.7	6.6	5.5	5.0	4.9	5.7	6.1	6.8	6.6
Precipitation	PrecAvg	48.8	3.5	3.1	4.3	4.2	3.9	4.2	4.0	4.4	4.6	4.2	4.0	4.4
	PrecMax	62.6	7.5	7.8	10.2	8.8	7.7	9.2	7.7	10.4	11.1	14.5	6.7	6.8
	PrecMin	35.6	1.8	1.4	0.8	1.7	0.9	1.1	1.5	1.9	1.3	0.4	1.6	1.5
	PrecStd	6.6	1.3	1.4	2.0	1.8	1.6	2.1	1.6	2.2	2.5	2.9	1.2	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	57.3	58.5	68.5	80.7	86.2	89.5	91.0	90.3	84.5	77.7	67.2	60.6
		MCWB	54.8	49.5	55.2	61.4	65.3	74.9	75.6	75.3	72.2	67.2	59.0	56.5
	2%	DB	53.9	51.9	60.7	72.0	80.8	84.4	87.7	85.5	80.7	72.3	62.6	53.9
		MCWB	51.6	46.1	49.0	56.1	64.4	70.8	74.4	74.0	68.8	64.9	57.3	50.2
	5%	DB	46.8	46.4	54.2	65.7	74.7	81.4	84.0	82.1	76.5	69.5	58.5	48.8
		MCWB	43.7	41.0	46.7	53.1	61.3	69.8	72.4	71.1	67.1	63.8	53.7	44.7
	10%	DB	41.9	43.1	48.9	61.0	70.1	77.9	81.5	80.5	73.1	64.7	54.8	45.1
		MCWB	37.9	37.9	42.8	49.7	58.9	66.9	70.6	69.7	65.3	59.3	49.5	40.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	56.0	54.3	57.6	63.9	69.5	76.1	78.3	77.8	74.6	70.6	62.2	56.6
		MCDB	56.7	56.1	66.9	75.6	79.8	85.6	87.1	85.6	80.1	74.4	64.8	58.4
	2%	WB	51.3	47.3	52.7	59.3	66.7	73.8	76.2	75.5	72.1	67.5	58.8	51.3
		MCDB	52.8	49.8	56.9	68.2	75.9	81.7	84.4	82.4	76.6	70.9	61.5	52.6
	5%	WB	44.6	42.6	47.4	55.5	63.7	71.7	74.4	73.8	70.0	64.2	54.9	45.9
		MCDB	46.5	45.5	52.5	62.0	72.1	78.8	81.1	79.5	73.5	67.1	57.8	48.1
	10%	WB	38.5	38.3	43.4	52.7	60.4	69.6	72.7	72.2	67.7	60.1	50.7	41.2
		MCDB	41.4	42.1	49.2	58.5	68.0	74.2	78.8	77.3	72.5	64.1	54.5	44.7
Mean Daily Temperature Range	5% DB	MDBR	14.4	15.2	16.4	18.4	19.5	18.1	18.3	17.7	17.8	17.8	15.2	13.7
		MCDBR	17.7	19.2	23.4	27.2	26.4	21.9	20.6	19.7	20.5	21.7	18.3	18.2
		MCWBR	16.4	15.6	16.2	15.8	14.1	11.5	10.4	10.1	11.6	13.8	15.1	16.0
	5% WB	MCDBR	17.5	18.2	21.7	22.4	23.0	18.7	17.3	16.4	16.6	17.2	16.6	16.3
		MCWBR	17.4	17.5	16.7	15.3	14.4	11.1	9.9	9.3	11.4	13.8	16.4	16.3
Clear-Sky Solar Irradiance	taub	0.297	0.308	0.331	0.359	0.403	0.449	0.465	0.440	0.371	0.338	0.311	0.296	
	taud	2.454	2.439	2.441	2.398	2.298	2.203	2.169	2.229	2.398	2.509	2.543	2.534	
	Ebn at Noon	272	287	292	289	277	264	258	261	274	272	266	265	
	Edh at Noon	24	29	32	36	41	45	46	43	33	27	22	21	
All-Sky Solar Radiation	RadAvg	586	860	1173	1508	1683	1834	1885	1666	1371	921	645	484	
	RadStd	45	77	90	161	165	174	104	101	118	91	53	47	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (70 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page